INCH-POUND
MIL-R-39007G
AMENDMENT 3
16 February 1996
SUPERSEDING
AMENDMENT 2
9 December 1993

MILITARY SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE),
ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-39007G, dated 16 May 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

TABLE II, Designation and Resistance columns, add "R100 to R988 inclusive" and "0.1 to 0.988 inclusive", respectively.

PAGE 5

* 2.1.1, delete "SPECIFICATIONS, MILITARY, MIL-R-39032" and its corresponding title.

* 2.1.1, STANDARDS, MILITARY, delete "MIL-STD-45662" and its corresponding title.

PAGE 6

* 2.1.2, add the following new publications:

"AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NCSL Z540-1 - Calibration Laboratory and Measuring and Test Equipment, General Requirements for .

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 (Standard specification for Electrodeposited Coating of Tin).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

"INTERNATIONAL ORGANIZATION for STANDARDS (ISO)

ISO 10012-1 - Quality Assurance Requirements for Measuring Equipment, Part 1:
Meterorological Confirmation System for Measuring Equipment.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)"

PAGE 8

Add new paragraph:

"3.5.1.5.3 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat effective 6 months from the date of this specification (see 6.9.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

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TABLE VII, delete and substitute:

"TABLE VII. Resistance temperature characteristic.

Resistance range	Resistance temperature characteristic parts per million/□C
0.1 □ R □ 0.499	±650
0.499 < R □ 1.0	±400
1.0 < R □ 10	±50
10 < R	±20

"

3.22.1, end of first sentence: Delete "±"0.5 percent" and substitute "±(0.5 percent +0.005 ohm)."

3.22.2, end of first sentence: Delete "±1.0 percent" and substitute "±(1.0 percent +0.005 ohm)."

3.23, delete "±0.5 percent" (two places) and substitute "±1 percent" (two places).

* TABLE VIII, delete and substitute:

"TABLE VIII. Reactance.

Style	Maximum effective series inductance - □H			Maximum effective parallel capacitance - pF
	< 50 ohms	□ 50 ohms		All resistance values
		< 3K ohms	□ 3K ohms	
RWR71N	0.20	0.60		1.5
RWR74N	0.50	1.20		1.5
RWR78N	0.65	5.00	12.00	1.5
RWR80N	0.20	0.37		1.5
RWR81N	0.20	0.37		1.5
RWR82N	0.20	0.37		1.5
RWR84N	0.50	1.20		1.5
RWR89N	0.20	0.60		1.5

"

Add the following new paragraph,

"3.26.4 Beryllium oxide (BeO). Manufacturers which use beryllium oxide in their construction shall mark each resistor body and resistor package with the symbol "BeO" "

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* 4.1.2, last sentence, delete "MIL-STD-45662" and substitute "ANSI/NCSL Z540-1, ISO 10012-1, or equivalent system as approved by the qualifying activity".

PAGE 16

4.4.4.2, add the following sentence: "In the event that the contractor meets or exceeds 100 ppm for ppm-2 the qualifying activity shall take the actions specified in EIA-554, method B."

PAGE 18

TABLE XII, Subgroup 2: Delete "4.7.4" and substitute "4.7.3".
Subgroup 3: Delete "Dielectric withstanding voltage" and all associated information.

4.6.1.2.1, first sentence, delete and substitute:

"Subgroup 1. Subgroup 1 tests shall be performed on a production lot basis on 100 percent of the product supplied under this specification."

PAGE 20

4.6.1.2.5.2, last sentence, delete and substitute:

"If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of the specification."

PAGE 21

TABLE XIV, delete and substitute:

"TABLE XIV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample
Resistance to solvents 1/	3.11	4.7.5	12

1/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency."

After 4.6.2.1.1.2, add new paragraph:

"4.6.2.1.1.2.1 Quarterly. Every 3 months, 10 sample units of each style and characteristic and of any resistance value shall be subjected to the tests specified in table XV, in the order shown."

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TABLE XV, delete and substitute:

" TABLE XV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample	Number of defects allowed
Monthly				
Subgroup 1				
Life	3.22	4.7.16	See 4.6.2.1.1.1	See 4.6.2.1.1.1
Subgroup 2			10	
Resistance temperature characteristic 1/	3.13	4.7.7	(6 highest	1
Moisture resistance	3.18	4.7.12	4 lowest)	
Quarterly 2/				
Dielectric withstanding voltage	3.16	4.7.10	10	
Insulation resistance	3.17	4.7.11	(5 highest	1
Low temperature storage	3.14	4.7.8	5 lowest)	
Terminal strength	3.19	4.7.13		
Semiannually 2/				
Subgroup 1				
Short time overload	3.15	4.7.9	30	
Thermal shock	3.12	4.7.6	(15 highest	1
Shock (specified pulse)	3.20	4.7.14	15 lowest)	
Vibration (high frequency)	3.21	4.7.15		
Subgroup 2 2/			(10 highest	
Reactance	3.25	4.7.19	30 10 3/	0
			10 lowest)	
Annual			102	
High temperature exposure	3.23	4.7.17	(51 highest 51 lowest)	1

1/ If the manufacturer can demonstrate that this test has been performed for five consecutive times with zero failures, the frequency of this test, with the approval of the qualifying activity, can be performed on an annual basis. If the design, material, construction or processing of the part is changed, or if there are any quality problems or failures, the qualifying activity may require resumption of the original test frequency.

2/ If the manufacturer can demonstrate that these tests have been performed for five consecutive times with zero failures, these tests, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform these tests every 3 years after the deletion as part of long term design verification. If the design, material, construction or processing of the part is changed, or if there are any problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirement in case of dispute.

3/ Ten samples of values representative of production in that period."

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* 4.6.3, delete in its entirety.

4.7.3a., delete in its entirety and substitute:

"a Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of the test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better."

PAGES 29 AND 30

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* 4.7.16f.(2), fourth line, delete "(10,000 hours)" and substitute "(10,000 +120, -0 hours)".

4.7.17c., delete "+275□C" and substitute "+250□C".

* 4.7.19 through 4.7.19.4, delete and substitute:

"4.7.19 Reactance (applicable to "N" and "Z" terminals and windings only) (see 3.25). Effective series inductance and parallel capacitance shall be determined using the method indicated (see 4.7.19.1). The units shall be mounted by their normal mounting means using the fixtures as suggested in the following procedure.

4.7.19.1 Method. The effective inductance of the resistors shall be measured using an HP4194A Impedance/Gain-Phase Analyzer with a 16047D fixture or an HP4192A LF Impedance Analyzer with a 16047A fixture or equivalent. The fixtures shall be adjusted so the lead length for all part sizes is minimum. The bridge is zeroed out with the appropriate shorting bar (see figure 3) prior to taking readings and every time the test clip spacing is changed."

* 5.1, delete and substitute:

"5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity."

PAGES 31 AND 32

* FIGURE 2, delete in its entirety.

PAGE 33

* FIGURE 3, ~~delete reference to fixtures~~ TF-A, ~~TF-B, and~~ TF-C.

PAGES 34, 35, AND 36

* FIGURES 4 and 5, delete in their entirety.

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FIGURE 6, delete and substitute:

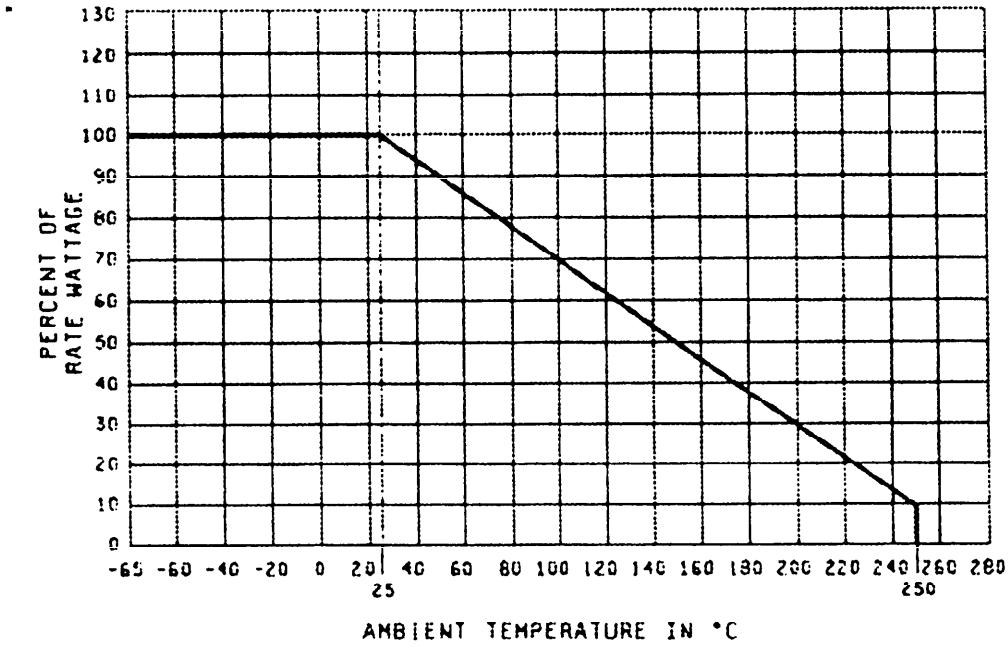


FIGURE 6. Derating curve for ambient temperatures. "

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6.7, add to end of last sentence, "and TABLE XVI-1 within".

Add new paragraph:

"6.9.1 Tin plated finishes. Tin plating is prohibited (see 3.5.1.5.3) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

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Add new TABLE XVI-1:

"TABLE XVI-1. Performance requirements

Style		RWR71 1/	RWR74 1/	RWR78	RWR80	RWR81	RWR84	RWR89	RWR82
Maximum resistance temperature characteristic in parts/million/□C-ppm (reference to 25□C)	0.1□ R □ 0.499	+650 ppm	+650 ppm	+650 ppm	+650 ppm	+650 ppm	+650 ppm	+650 ppm	+650 ppm
	0.499 < R □ 1.0	+400 ppm	+400 ppm	+400 ppm	+400 ppm	+400 ppm	+400 ppm	+400 ppm	+400 ppm
	1.0 < R □ 10	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm	±50 ppm
	10 < R	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm	±20 ppm
Maximum ambient temperature at rated wattage (see figure 6)		25□C	25□C	25□C	25□C	25□C	25□C	25□C	25□C
Maximum ambient temperature at zero wattage derating (see figure 6)		275□C	275□C	275□C	275□C	275□C	275□C	275□C	275□C
Power rating in watts		2	5	10	2	1	7	3	1.5
Maximum percent change in resistance ± 2/	Conditioning (see 3.8)	.2	.2	.2	.2	.2	.2	.2	.2
	Thermal shock (see 3.12)	.2	.2	.2	.2	.2	.2	.2	.2
	Low temperature storage (see 3.14)	.1	.1	.1	.1	.1	.1	.1	.1
	Short-time overload (see 3.15)	.2	.2	.2	.2	.2	.2	.2	.2
	Dielectric withstanding voltage (see 3.16)	.1	.1	.1	.1	.1	.1	.1	.1
	Moisture resistance (see 3.18)	.2	.2	.2	.2	.2	.2	.2	.2
	Shock (specified pulse) (see 3.20)	.1	.1	.1	.1	.1	.1	.1	.1
	Vibration, high frequency (see 3.21)	.1	.1	.1	.1	.1	.1	.1	.1
	Life (see 3.22)	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22	see 3.22
	High temperature exposure (see 3.23)	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23	see 3.23
	Insulation resistance (see 3.17)	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min	1,000 megohms min
	Insulation resistance (after moisture resistance) (see 3.18)	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min	100 megohms min
Resistance tolerances ±percent (see table IV)		1, .5, .1	1, .5, .1	1, .5, .1	1, .5, .1	1, .5, .1	1, .5, .1	1, .5, .1	1, .5, .1

1/ Not for use on new design; for replacement purposes only.
2/ Where total resistance change is 1 percent or less, it shall be considered as ±(__ percent +0.005 ohm)."

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done for convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 17, 19, 80, 85
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1428)

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